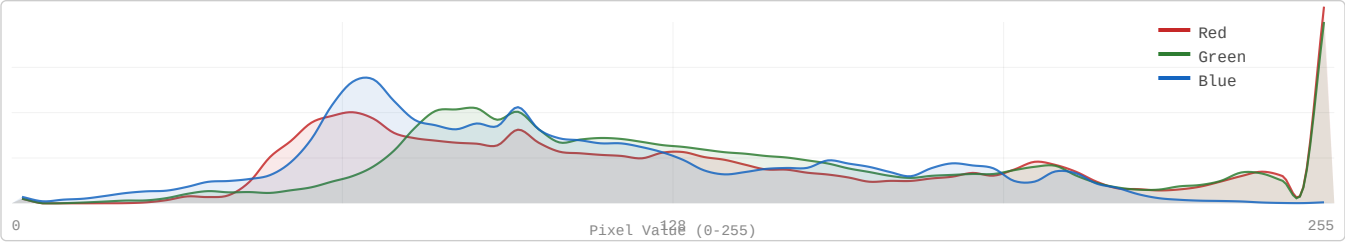
Decorrelation Gap: KODIM06

768x512 | 24-bit RGB | PNG (lossless)

Sailboat at Anchor | Near-degenerate (PC1 = 98.42%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9772	0.9603
G	0.9772	1.0000	0.9911
B	0.9603	0.9911	1.0000

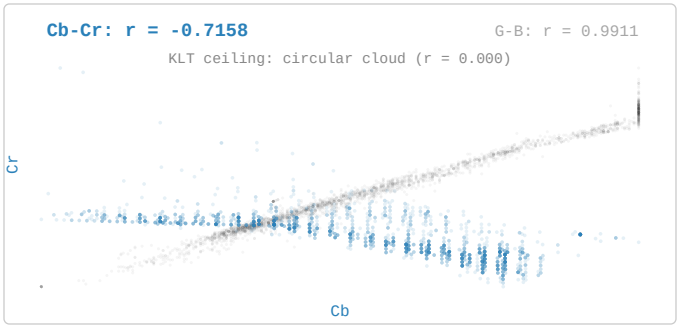
Cb-Cr: -0.716

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.8502	0.4413
Cb	-0.8502	1.0000	-0.7158
Cr	0.4413	-0.7158	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.977168	-0.850217	0.8502
R-B → Y-Cr	0.960329	0.441289	0.4413
G-B → Cb-Cr	0.991059	-0.715758	0.7158
Avg |r|	0.976185	0.669088	0.6691

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

31.5% Low

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

2.4%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	46.26 dB
G	50.15 dB
B	42.44 dB
Overall	45.21 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

PC1	98.42%
PC2	1.40%
PC3	0.18%

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6462	-0.5960	-0.4766	10128.67
PC2	-0.7417	0.3436	0.5760	143.68
PC3	0.1796	-0.7257	0.6642	18.67

8. PCA Derived Metrics

Condition Number

542.58

PC1/PC2: 70.50 | PC2/PC3: 7.70

Blue Channel Independence

2.4%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.65)

9. Redundancy Profile

Classification: Near-degenerate | CN: 542.58 | PC1: 98.4%

RGB Correlation: Avg |r| = 0.976185 | R-G: 0.9772 | R-B: 0.9603 | G-B: 0.9911

YCbCr Residual: Avg |r| = 0.669088 | Y-Cb: -0.8502 | Y-Cr: 0.4413 | Cb-Cr: -0.7158

Efficiency: 31.5% of KLT ceiling | Gap: 0.6691 avg |r| remaining

Blue Channel: 2.4% independent | 4:2:0 PSNR: 45.21 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.